R2_DEPI2_CAI2_ISS4_S1_Cisco Network Administrator

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Project Planning & Management for

Developing a Smart Office Network Solution

Supervisor

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Project overview

Design and implement a secure, efficient network for a mid-sized smart office with approximately 100 employees. The network must support IoT devices (smart lighting, cameras, environmental sensors), employee workstations, printers, and remote access capabilities.

Objective: Provide secure, scalable, and efficient networking for IoT devices, employee workstations, and guest access.

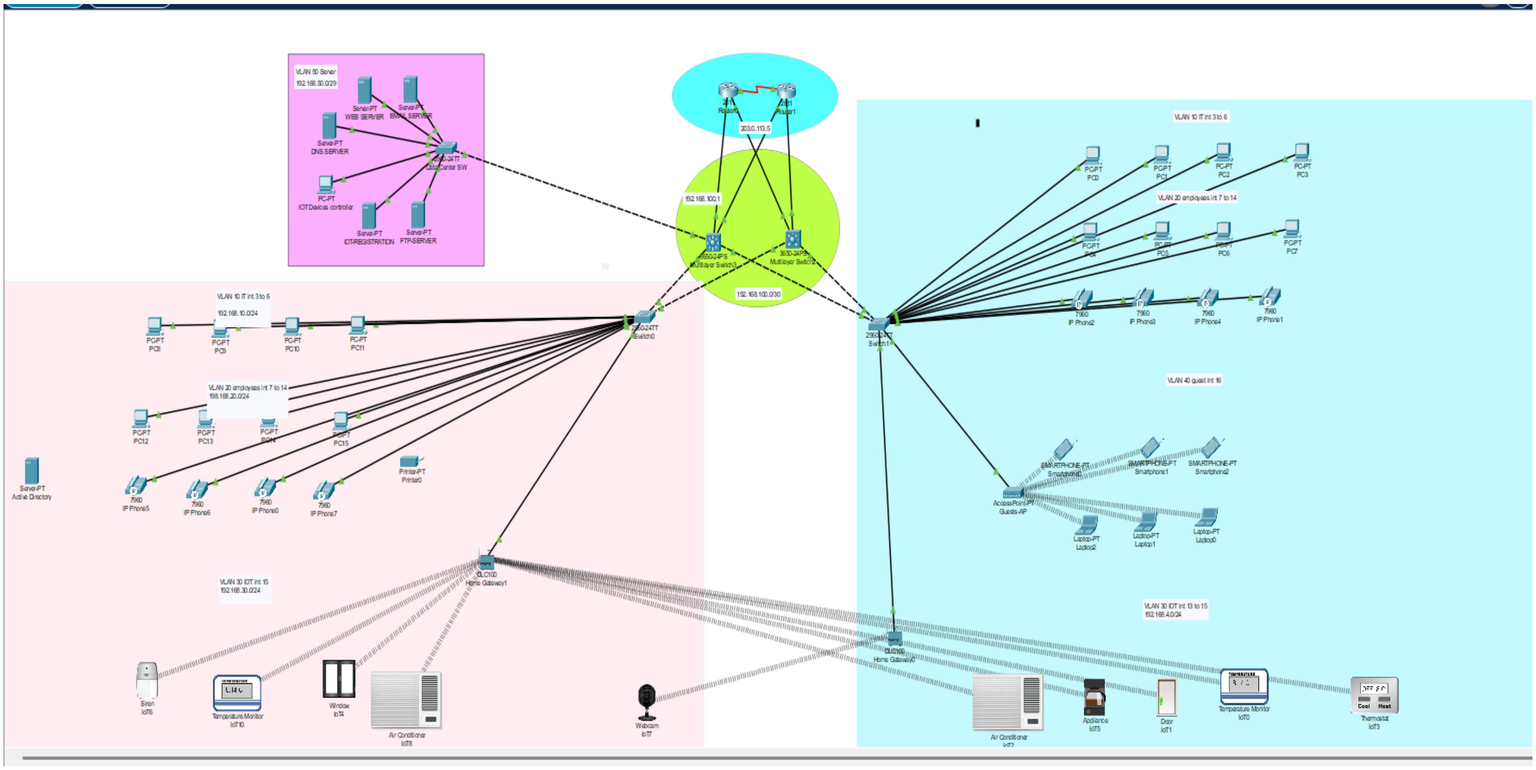
project Features:

- IoT integration (e.g., smart lighting, AC, sensors)
- VLAN segmentation
- Secure remote access
- Quality of Service (QoS)

The project cover:

- Network Infrastructure Design – Planning and implementing a scalable network.
- IoT Integration – Connecting and managing smart devices securely.
- Remote Access & Monitoring – Enabling authorized users to monitor and control devices remotely.
- Scalability & Future Expansion – Ensuring the network can handle additional devices and users.

Network Topology



Description:

- **Devices:** Routers, multilayer switches, access switches, PCs, IP phones, printers, IoT sensors, wireless access points.
- **Core Routing:** Router-on-a-stick configuration using sub-interfaces on the multilayer switch.
- **WAN Connection:** Connected to ISP router with IP 230.1.1.5.
- **Key IP Ranges:**
- VLAN 10 (IT): 192.168.10.0/24
- VLAN 20 (Employees): 192.168.20.0/24
- VLAN 30 (IoT): 192.168.30.0/24
- VLAN 40 (Guest): 192.168.40.0/24
- VLAN 99 (Management): 192.168.100.0/24

VLAN Assignments

VLAN ID	Name	Purpose	IP Range
10	IT	Servers and IT PCs	192.168.10.0/24
20	Employees	Office PCs & Phones	192.168.20.0/24
30	IoT	Smart Devices	192.168.30.0/24
40	Guest	Smartphones & Laptops	192.168.40.0/24
50	Server-room	Servers and controller	192.168.50.0/28
99	Management	Admin Network Devices	192.168.100.0/24

CONFIGURE STEPS

0. Network Design and beatification.

1. Basic settings to all devices plus ssh on the routers and switches.

Configure hostnames, passwords, banners, and enable secure remote access using SSH for device management.

2. VLANs assignment plus all access and trunk ports on switches.

VLANs and assign access ports to end devices. Configure trunk ports to carry multiple VLANs between switches.

3. Switchport security to server-side site department.

port security on server-side switches to limit MAC addresses and prevent unauthorized access.

4. Subnetting and IP addressing

IP ranges per VLAN using subnetting and assign IPs to interfaces and critical devices.

5. OSPF on the routers and switches.

OSPF routing to enable dynamic route sharing between network segments.

6. Static IP address to server Room devices.

fixed IP addresses to servers and infrastructure devices for reliable connectivity.

7. DHCP server device configurations.

DHCP to dynamically assign IP settings to client devices within each VLAN.

8. Inter-VLAN routing on the switches plus IP DHCP helper addresses.

Enable routing between VLANs on L3 switches and configure DHCP relay to forward requests to the DHCP server.

9. Wireless network configurations.

Configure access points, SSIDs, VLAN tagging, and wireless security settings

10. Site-to-site IPSec VPN

Establish a secure VPN tunnel between remote sites for encrypted data transmission.

11. Default static route

Define a default route to forward traffic destined for external networks to the next-hop gateway.

12.NAT and Access Control List

Enable NAT for internet access using a single public IP and apply ACLs to control traffic based on security rules.

13.Verifying and testing configurations.

Test and validate all configurations using commands and tools to ensure full network functionality.

Security Policies

Port Security:

Enabled on server-side switches with MAC address limits and violation shutdown.

SSH Access:

Only secure remote management allowed via SSH on routers and L3 switches.

ACLs:

Configured on router to restrict guest VLAN access and permit only essential traffic.

NAT:

enabled for internal devices to access the internet.

VPN:

IPSec VPN tunnel established for secure site-to-site communication.

DHCP Snooping:

on access switches to prevent rogue DHCP servers.

Enable Secret:

Encrypted enable password set on all network devices